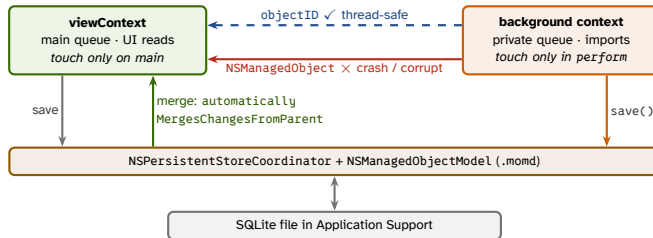


In one line: Secrets → **Keychain**; small settings → **UserDefaults**; blobs → **files** in the right sandbox directory; a queryable object graph → **Core Data** / **SwiftData**; recomputable in-memory → **NSCache**. In Core Data every `NSManagedObjectContext` is **bound to one queue**: touch it and its objects only inside `perform`, and hand `NSManagedObjectIDs` — never objects — across queues.

Core Data — `NSPersistentContainer`



App sandbox (deleted on uninstall)

Documents/ backed up · user files (Files app)
 Library/Application Support/ backed up · app data, DB
 Library/Caches/ **not backed up** · OS may purge
 Library/Preferences/ backed up · UserDefaults plist
 tmp/ **not backed up** · purged when not running
 Keychain lives *outside* the sandbox → can outlive the app

Decision table

Store	For	Secrets	Backup	Uninstall
UserDefaults	small prefs, flags (whole plist read into RAM)	no	yes	deleted
Keychain	tokens, passwords, keys	yes	encrypted backups	survives*
Files	images, downloads, any size	no	per dir	deleted
Core Data	10k+ related, queried records	no	yes	deleted
SwiftData	same, iOS 17+, SwiftUI-first	no	yes	deleted
NSCache	recomputable; RAM, self-evicting	no	—	on quit

*in practice, not a documented guarantee. Files get Data Protection

(.completeFileProtection), but that is not app-level secret storage. NSCache: thread-safe, evicts under memory pressure, does not copy keys; limits are hints.

Core Data

- `viewContext = main queue`. `newBackgroundContext() / performBackgroundTask { ctx in }` = **private queue**.
- `perform (async; await-able iOS 15) / performAndWait` (blocks caller) run the block *on the context's queue* — the only legal access, even a read or a relationship (fires a fault).
- `objectID` is *temporary* until `save`; `object(with:)` returns a fault, `existingObject(with:)` throws if the row is gone.
- Merge**: contexts do not see each other's saves. `viewContext.automaticallyMergesChangesFromParent = true` (picks up sibling saves via the coordinator) or `mergeChanges(fromContextDidSave:)` inside the target's `perform`. FRC / `@FetchRequest` watch only their own context.
- Conflict policy default = **error** (save throws); usual choice: `NSMergeByPropertyObjectTrumpMergePolicy`.
- Parent/child**: `child.parent = viewContext`; `child save()` only pushes up *in memory* — disk needs the parent's save.
- Faulting**: fetched objects are placeholders filled on access. N+1 fix: `relationshipKeyPathsForPrefetching`, `fetchBatchSize`.
- Batch** requests bypass contexts → merge returned IDs.
- Migration**: *lightweight* = inferred (add/remove, optional/default, rename via **Renaming ID**), on by default; *heavyweight* = `NSMappingModel + NSEntityMigrationPolicy`.

Keychain — `kSecAttrAccessible...`

`WhenUnlocked` = **default**, only while unlocked ·

`AfterFirstUnlock` = after first unlock since boot — needed for **background** work · `WhenPasscodeSetThisDeviceOnly` = needs a passcode · suffix `ThisDeviceOnly` = never restored to another device. `SecItemAdd` twice → `errSecDuplicateItem` (use `SecItemUpdate`).

Example — import off main, show on main

```
let c = NSPersistentContainer(name: "Model")
c.loadPersistentStores { _, e in precondition(e == nil) }
c.viewContext.automaticallyMergesChangesFromParent = true

func importItem(_ dto: ItemDTO) async throws -> NSManagedObjectID {
    let bg = c.newBackgroundContext() // private queue
    return try await bg.perform { // on bg's queue
        let item = Item(context: bg); item.title = dto.title
        try bg.save() // -> coordinator
        return item.objectID }
    }
    // on the main actor:
    let id = try await importItem(dto)
    let item = try c.viewContext.existingObject(with: id) as! Item
```

SwiftData (iOS 17+)

`@Model` class ≈ entity · `ModelContainer` ≈ container · `ModelContext` ≈ context (mainContext is `@MainActor`, autosaves) · `@Query(sort: \Trip.date)` ≈ live FRC in a view. Background: a `@ModelActor` owns its context; pass `PersistentIdentifier`, never the `@Model`. Migrations: `VersionedSchema + SchemaMigrationPlan`.

Traps

- ! A managed object on the wrong queue “works” until it crashes. Launch arg `-com.apple.CoreData.ConcurrencyDebug 1` traps the first violation.
- ! Background save, UI stale → not merged into `viewContext`.
- ! Rename without a `Renaming ID` = drop + add → data lost.
- ! Token in `UserDefaults` = readable from a backup. In `Keychain` with `WhenUnlocked` = unreadable during background refresh.
- ! `Keychain` survives reinstall: clear stale tokens on first run.
- ! Re-downloadable files in Documents bloat iCloud backup → Caches.

Keychain item classes (`kSecClass...`)

GenericPassword	token, API key — keyed by service + account
InternetPassword	password for a server — keyed by server, protocol, port
Certificate · Key	X.509 certificate · private/public key
Identity	certificate + its private key (client TLS)

Remember

Secret → **Keychain**, **setting** → **UserDefaults**, **blob** → **file**, **graph** → **Core Data** / **SwiftData**, **rebuildable** → **NSCache**. One context, one queue; pass IDs, not objects.